

Online Retail Dataset

Customer Segmentation & RFM Analysis

This analysis aims to support data-driven decision making in e-commerce by identifying key customer segments and revenue patterns.

1. Project Objective

The goal of this project is to analyze customer purchasing behavior using the Online Retail dataset and generate actionable business insights.

The analysis includes:

- Data cleaning and preprocessing
- Exploratory SQL analysis
- RFM scoring
- Customer segmentation
- Business recommendations

2. Data Cleaning

The dataset was cleaned and prepared using Python (Pandas):

- Removed rows with missing CustomerID
- Converted InvoiceDate to datetime format
- Created TotalPrice = Quantity × UnitPrice
- Identified negative quantities (returns)
- Ensured consistent data types for SQL queries

3. SQL Analysis (DuckDB)

Using DuckDB, several exploratory analyses were performed:

Top Revenue-Generating Products

```
SELECT Description, SUM(Quantity) AS TotalSold  
  
FROM df  
  
GROUP BY Description  
  
ORDER BY TotalSold DESC  
  
LIMIT 10;
```

Countries with the Highest Number of Orders

```
SELECT Country, COUNT(DISTINCT InvoiceNo) AS TotalOrders

FROM df

GROUP BY Country

ORDER BY TotalOrders DESC;
```

Monthly Revenue Trends

```
SELECT DATE_TRUNC('month', InvoiceDate) AS Month,

SUM(Quantity * UnitPrice) AS MonthlyRevenue

FROM df

GROUP BY Month

ORDER BY Month;
```

Top Spending Customers

```
SELECT CustomerID,

SUM(Quantity * UnitPrice) AS TotalSpent

FROM df

GROUP BY CustomerID

ORDER BY TotalSpent DESC

LIMIT 10;
```

Most Frequently Purchased Product

```
SELECT Description,

SUM(Quantity) AS TotalQuantity

FROM df

GROUP BY Description

ORDER BY TotalQuantity DESC

LIMIT 1;
```

These queries helped uncover key patterns in customer and product behavior.

4. RFM Analysis

For each customer, the following metrics were calculated:

- **Recency** — Days since the last purchase
- **Frequency** — Number of unique invoices
- **Monetary** — Total spending

RFM scores were generated using quantile-based ranking, and each customer received a 3-digit score (e.g., 444, 321, 112).

5. Customer Segmentation & Business Insights

Champions

- Highest-value customers
- Frequent purchases and high spending
- **Strategy:** VIP programs, exclusive offers, early access campaigns

Loyal Customers

- Consistent buyers with strong engagement
- **Strategy:** Loyalty rewards, personalized product recommendations

Potential Loyalists

- New customers with high potential
- **Strategy:** Welcome discounts, onboarding campaigns

At Risk

Previously valuable customers who haven't purchased recently

Strategy: Win-back emails, targeted promotions, limited-time discounts

Hibernating

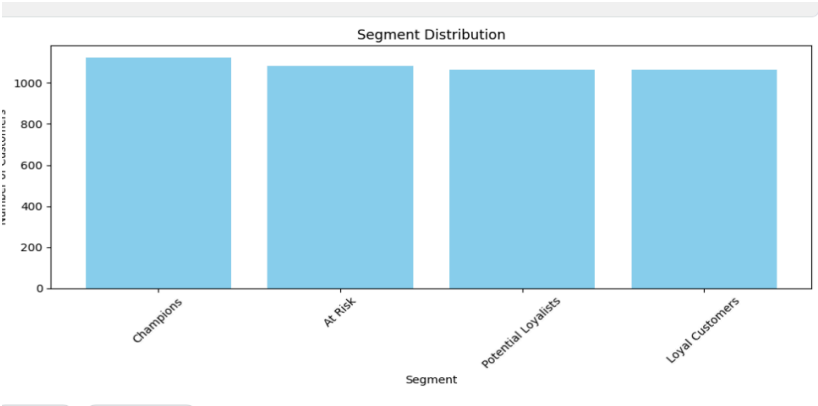
- Long inactive and low-value customers
- **Strategy:** Re-activation campaigns, seasonal offers

These segments help businesses allocate marketing resources more effectively.

6. Visualizations

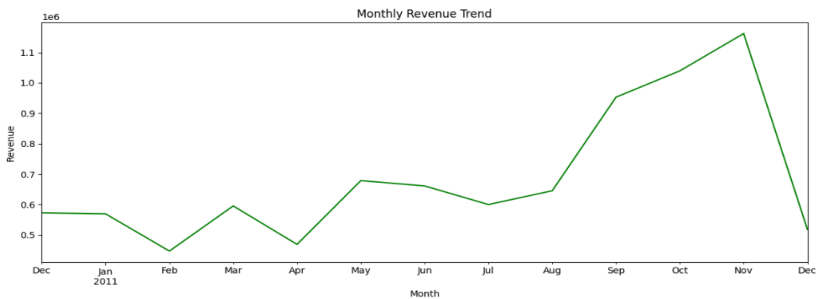
Several visualizations were created to support the analysis:

Segment distribution bar chart



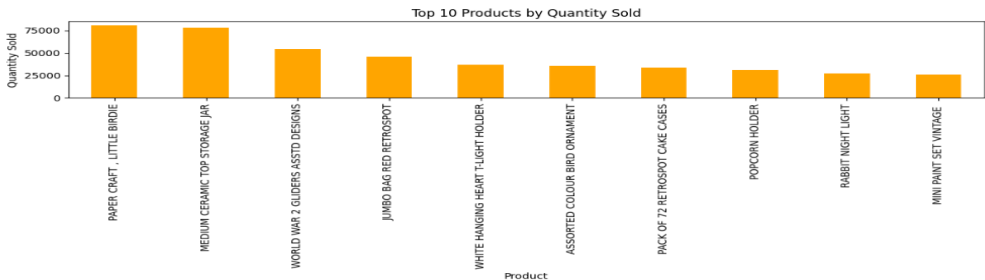
Explanation: This chart shows the distribution of customers across RFM-based segments. *Champions* form the largest group, reflecting strong engagement and high spending. The notable size of the *At Risk* segment indicates potential churn and highlights the need for targeted retention strategies.

Monthly revenue time series



Explanation: The time series reveals clear seasonal revenue patterns, with a significant increase during the final quarter of 2011. This suggests strong holiday-driven demand and indicates opportunities for seasonal promotions and inventory planning.

Top 10 products by quantity sold



Explanation: The chart shows that a small number of products account for a disproportionately large share of total sales. This concentration suggests that these top-selling items should be prioritized in inventory management, marketing campaigns, and stock availability planning.

7. Conclusions & Recommendations

Conclusions

The analysis highlights several key business insights:

- A small group of high-value customers (Champions) drives a large share of revenue.
- A notable portion of customers is at risk of churn, requiring attention.
- Revenue shows strong seasonal peaks, especially in Q4.
- Sales are concentrated in a limited number of top-performing products.

These findings provide a clear view of customer value, purchasing patterns, and revenue drivers.

Recommendations

- **Prioritize retention** for Champions and Loyal Customers through exclusive offers and personalized engagement.
- **Target At Risk customers** with win-back campaigns and tailored incentives.
- **Optimize inventory** by ensuring consistent availability of top-selling products.
- **Support Potential Loyalists** with onboarding promotions to increase repeat purchases.
- **Leverage seasonal demand** by planning marketing and stock strategies around peak periods.

8. Technologies Used

- Python (Pandas, Matplotlib)
- DuckDB (SQL)
- Jupyter Notebook
- Data visualization
- RFM segmentation methodology

Overall, this project demonstrates the value of combining SQL, Python, and RFM methodology to uncover actionable customer insights in retail analytics.

Appendix: Full Code & Notebook

The full Python, SQL, and visualization code used in this analysis is available in the Kaggle Notebook and GitHub repository below:

Kaggle Notebook:

<https://www.kaggle.com/code/gulayaksoy/online-retail-rfm-analysis>

GitHub Repository:

<https://github.com/Gulaksoy/online-retail-rfm-analysis>

This repository contains:

- The full Jupyter Notebook (.ipynb)
- All data cleaning and preprocessing steps
- SQL queries executed in DuckDB
- RFM scoring code
- Visualization generation code
- Additional exploratory analysis